

Floruvi farm plan, Bengaluru

Feasibility brief · version 1.1 · 29 September 2026. Adds your ₹3,000 tower plan, shared pumps, raft beds, and two reviewers' fixes to your cost figures. Built from 18 research agents, one model and 2 reviewers (724 unique sources). Version 2 follows after the final model check.

The answer

600 towers now: not affordable. Bought kits cost about ₹1.49 crore; your own towers about ₹1.19 crore. You have ₹20 lakh. **200 → 6,000 towers in 6 months: not possible.**

Your ₹3,000 tower: the plastic can cost that (about ₹1,630 a tower), but the moulds cost ₹12.4–26 lakh. At 200 towers your own towers cost the same as bought kits (₹50.9 lakh). They start to save money above about 600 towers. The ₹3,000 body is realistic above about 2,200 towers, because plastic now costs ₹154.7/kg.

Towers lose money when wages, water and power are legal and real: ~~₹17,714~~ a month at 200 own towers. Your figures show +₹49,190 only because the guard, caretaker, water, power and repairs are below legal or physical minimums.

Best option found: floating raft beds (DWC). 200 bed units (18,000 plants) cost about ₹42.9 lakh and make about +₹86,990 a month with legal, realistic costs (+₹1,14,460 per 40 days). This is an estimate: pilot it first.

No growing profit by December 2026. Microgreens and trading greens bring first cash in 2–3 weeks.

Key numbers (base case)

Metric	Value
Build cost, 200 units: bought at ₹17,000 / kits / your own towers / raft beds	₹61.6 lakh / ₹50.9 lakh / ₹50.9 lakh / ₹42.9 lakh
Profit a month, legal and realistic: 200 own towers / 200 raft-bed units	₹17,714 / +₹86,990
Profit per 40 days, legal and realistic: 200 raft-bed units	+₹1,14,460
Raft beds: loan → EMI, and profit left after the EMI	₹35.9 lakh → ₹59,524 a month; +₹27,466 left
Moulds for your own tower	ring only ₹12.4 lakh (₹5.9–24.7 lakh); full Indian set ₹26 lakh
Plastic (PP) price, IOCL Bangalore	₹90.7/kg on 1 Jan 2026 → ₹154.7/kg on 16 Sep 2026
Shared pumps (10 towers per 1,000 L tank)	10.5 W per tower instead of 20 W; about ₹2,970 per tower
What buyers take from a new seller at good prices	95 kg a week by month 3; 400 kg a week by month 12
First cash	trading greens about 14 days · microgreens about 19 days · first harvest about 9 weeks

Three cost views. *Your figures:* guard ₹10,000, helpers ₹40,000, water ₹10,000, power ₹10,000, caretaker ₹15,000; repairs, grower, accounting, insurance and marketing ₹0 (plus RO filter parts and the FSSAI fee). *Legal and realistic:* you grow up to 200 units; the caretaker covers nights (no separate guard up to 200 units); caretaker at the legal 7-day rate (about ₹22,300 with extras); real tanker water, power, repairs, insurance and samples; labour 29% higher than version 1. *Research costs:* a paid grower and an agency guard above 150 units. Profit = cash profit before loan payments and tax, at month-12 demand. Raft-bed numbers are estimates from one research agent (140 g heads, 25% less labour).

Scorecard: will this work?

Why it can work

Metric	Value	Source
Raft beds make money with legal costs	200 units: +₹86,990 a month; profit covers the loan EMI	Model + systems research (estimate)
Top crops earn money per tower	Arugula ₹1,246, Lollo Rosso ₹845 a month after seeds, packs, delivery and waste	Model, Hyperpure prices
Heavier heads outside towers	Raft beds about 140 g against 90 g in towers; towers 31% lighter in a controlled study	Touliatos 2016; systems research
Raft beds survive power cuts	12–72 hours before damage (estimate), against 30–60 minutes for towers	Systems research
Shared pumps cut power	10 towers per tank: 10.5 W per tower; saves ₹25,000–34,000 a month at 600 towers	Systems research
Mild climate, no land rent	Nov–Feb ideal; owned 1 acre fits about 1,500–1,600 towers or about 200 bed units with room to spare	IMD; owner
Real demand	About 400 kg a week at good prices by month 12 (about 40 business accounts)	Demand research
Fast side-line cash	Microgreens ₹1,600–2,500/kg in shops; break-even about ₹700/kg; first sales about day 15	BigBasket; model
Cheap long-term loans	AIF: about 6%, up to ₹2 crore, up to 2 years before repayments, no collateral	AIF guidelines

Why it may not work

Metric	Value	Source
Towers lose money with legal costs	200 own towers –₹17,714 a month; 600 own towers –₹2,68,203	Model, reviewers
Your cost figures break the law or physics	₹10,000 guard and ₹15,000 7-day caretaker are below minimum wage; ₹10,000 water and power fit about 64–79 towers	Reviewers; labour research
Capital gap	Raft beds 200 units ₹42.9 lakh; own towers 200 ₹50.9 lakh. You have ₹20 lakh	Model
Moulds before the first tower	₹12.4–26 lakh; the ₹3,000 body is realistic only above about 2,200 towers	Moulding research
Plastic price shock	PP up 70% in 2026 (₹90.7 → ₹154.7/kg); ₹30/kg moves the body cost about ₹300	IOCL price lists
Buyers are the limit	400 kg a week at good prices by month 12; 600 towers harvest about 640 kg a week	Demand research
Early losses and heat	20–35% losses at first; March–May lettuce –40%; root rot can kill a shared tank	Climate research; Univ. of Kentucky
Water and power	Tankers ₹2,000–2,850 a load in Feb–May; inverters last 3–4 h, BESCOM shutdowns 4–8 h	Water and power research; reviewers
Strong competitors, big failures	BigBasket 24 hydroponic lettuces at ₹47–85; Deep Rooted lost ₹52 crore and shut (2025)	Price and demand research

How to make it a successful business

Action	Effect	When
Pilot raft beds next to one 10-tower cluster before buying more towers or moulds	Raft beds: +₹1,04,704 a month more than own towers at 200 units (estimate)	Oct 2026
Sell before you grow: signed weekly orders; add units after 4 weeks at 70%+ sold at good prices	Keeps sales near 85% at good prices; at 600 towers this is worth about ₹89,000 a month	Now
Make your figures legal: one resident caretaker at the legal rate covers nights up to 200 units	Avoids a ₹33,000–75,000 guard; removes the legal risk	Now
Raise your average price 20% (Floruvi boxes, chefs)	+₹40,955 a month at 200 towers; +₹91,954 at 600	Month 1 on
Shared pumps: 10 towers per 1,000 L tank, standby pump, float and SMS alarm	Half the pump power; ₹25,000–34,000 a month saved at 600 towers	With the pilot
No moulds for the first 200 towers. Then mould only the ring in India; get 3 Indian and 2 Chinese written quotes	Saves ₹12.4–26 lakh until volume justifies it; about ₹5,600 saved per tower above 600	After the pilot
Check design rights before tooling (IP India design and patent search)	Avoids a claim of up to ₹50,000 per design, or damages and an injunction	Before moulds
Fix water: roof rainwater + Cauvery or a permitted borewell	About ₹22,000 a month saved at 200 towers against tankers	Jan–Mar 2027
Quick cash: microgreens 100 trays a week and trading greens	+₹16,500–24,500 and +₹7,600–30,000 a month	Oct 2026
Fund growth with an AIF loan after 2–3 good crop cycles	Raft beds 200 units: EMI ₹59,524 against ₹86,990 profit	2027

Profit three ways: your figures, legal and realistic, research

Base case	200 own towers	200 raft-bed units (DWC)	600 own towers
Plants	18,000	18,000	54,000
Build cost (own towers include ₹12.4 lakh of moulds)	₹50.9 lakh	₹42.9 lakh	₹1.19 crore
Harvest a week / share sold at a good price	223 kg / 85%	334 kg / 85%	641 kg / 62%
Running costs a month: your figures	₹91,153	₹94,483	₹1,03,660
Running costs a month: legal and realistic	₹1,58,058	₹1,43,412	₹5,41,961
Profit a month: your figures	+₹49,190	+₹1,35,918	+₹1,70,098
Profit a month: legal and realistic	–₹17,714	+₹86,990	–₹2,68,203
Profit per 40 days: legal and realistic	–₹23,308	+₹1,14,460	–₹3,52,899
Profit a month: research costs	–₹1,38,426	–₹33,722	–₹3,18,500
Loan → EMI (6%, 6 years), legal and realistic	₹44.8 lakh → ₹74,208	₹35.9 lakh → ₹59,524	₹1.36 crore → ₹2,26,070
Profit left after the EMI (and after ₹40,000 until Aug 2027)	–₹91,922 (–₹1,31,922)	+₹27,466 (–₹12,534)	–₹4,94,274 (–₹5,34,274)

Range, legal and realistic, bad to good season and prices: raft beds 200 units –₹1,64,593 to +₹3,49,087 a month; own towers 200 –₹2,36,004 to +₹1,87,170. All numbers are month-12 steady state, not month 3. Raft beds need about 1,120 m² under cover, which fits on your 40-cent plot (1,619 m²).

Where your cost figures hold, and where they break

Your figure	Realistic up to	Research and the law
Caretaker ₹15,000	6 days × 8 hours only	Live-in 7-day work: legal minimum ₹18,699 a month (agriculture rate); about ₹22,300 with extras
Night guard ₹10,000	Not legal at any size	Legal 12-hour post about ₹33,000 direct, ₹61,000–75,000 through an agency. Up to 200 units, the caretaker covers nights with CCTV and a siren
Helpers ₹40,000	About 140–190 towers	Labour is 29% higher than in version 1 (49 hours per 1,000 plants per 40 days)
Grower ₹0 (you)	About 150–200 units	You also sell and deliver; add a part-time sales person after about 10 accounts
Water ₹10,000	About 64–70 towers	200 towers need about ₹30,796 of tanker water; Mar–May up to 1.5–2×
Power ₹10,000	About 79 towers (150 with clusters)	200 own towers in clusters about ₹13,312; with one pump per tower about ₹23,212
Repairs ₹0	First 6 months	About ₹75 per tower a month, plus 1.5% of the structure a year
Insurance ₹0	Not with a loan	About ₹1,900 a month at 200 towers; lenders require it
Accounting, marketing ₹0 (AI)	Mostly	FSSAI licence about ₹625 a month; samples and sales travel

Build it cheaper: your ₹3,000 tower

Your 5 towers cost ₹17,000 each delivered, each with its own pump. You plan to copy them in plastic, about 6 parts: net cup, ring panel, seed holder and basin. The research priced the plastic, the moulds and the other parts.

Item	Value (low / base / high)
Plastic per 88-site tower	5.5 / 8.1 / 11.4 kg → moulded parts ₹720 / 1,630 / 2,790 before GST
PP price, IOCL Bangalore (before GST)	₹90.7/kg on 1 Jan 2026 → ₹154.7/kg on 16 Sep 2026 (West Asia conflict)
Indian steel (P20) moulds	net cup ₹1.2 lakh · ring ₹4 / 8 / 15 lakh · basin ₹3 / 6 / 10 lakh (needs a 600–800 t press) · full set ₹9.5 / 19 / 34 lakh; 4–12 weeks
Chinese moulds	USD 15,000 / 27,000 / 46,000 for the set, plus 2–3 weeks by sea and import duty
Aluminium moulds	Last about 10,000 shots: enough for about 600 towers
Your own machines	80–160 t ₹6–18.5 lakh; 450 t new ₹52 lakh. Worth it only at 20,000–37,000 towers a year
Design work	Freelancers ₹500–2,000+/hour; Bengaluru agency ₹3,500–4,500/hour; about 60–160 hours; design search ₹15,000–30,000
Costs the ₹3,000 leaves out	Net cups ₹176–396, pipe and fittings ₹120–350, pump/timer/tank share ₹450–2,000, assembly, freight, 2–8% rejects, 18% GST

Complete cost per tower by route (base, GST included; tooling spread over the volume)

Route	50	200	600	6,000
India steel moulds, all parts	₹55,808	₹16,703	₹8,013	₹4,102
India aluminium moulds	₹31,028	₹10,508	₹5,948	₹4,309
China moulds, moulded in India	₹83,614	₹23,654	₹10,330	₹4,334
China parts imported	₹68,916	₹22,177	₹12,153	₹6,292
Rotomoulded basin + thermoformed rings	₹19,773	₹9,519	₹7,240	₹6,215
Ready kits + GST	₹12,940	₹12,320	₹11,701	₹11,082
DIY PVC towers	₹5,545	₹5,545	₹5,545	₹5,545
Ring mould only (buy tub, cups)	₹28,779	₹10,119	₹5,972	₹4,106
Ring mould only + shared cluster tank	₹28,907	₹10,247	₹6,100	₹4,234

Cheapest: DIY PVC at 50–600 towers; for your own design, the rotomoulded route at 50–200 and aluminium or ring-only moulds at 600; Indian injection moulding only near 6,000. The ₹3,000 body is realistic from about 2,200 towers (ring-only route) or 7,900 (full steel set); from about 350 towers if plastic falls back near ₹100/kg. Tooling budgets with GST and development: ring only ₹12.4 lakh, full Indian steel ₹26 lakh, aluminium ₹13.7 lakh, China landed about ₹40 lakh, thermoforming plus rotomoulding ₹6.8 lakh.

Advice: build no moulds for the first 200 towers. Tie any plastic price in a contract to the IOCL list. Do not buy machines. Search IP India designs and patents (or pay a patent agent) before tooling: copying a registered design risks a claim of up to ₹50,000 per design, or damages and an injunction.

Shared pumps, and a better system than towers

Towers per tank	Pump power per tower	kWh per tower a month	Tank	Capex per tower	Plants at risk if it fails
1	20.0 W	9.6	30 L	kit	90
6	14.2 W	6.8	500 L	₹3,091	540
10	10.5 W	5.1	1,000 L	₹2,970	900
20	10.5 W	5.1	2,000 L	₹2,380	1,800

Recommended: 10 towers per buried or insulated 1,000 L tank, one 105 W pump (SOBO WP-7000 class) plus a standby pump, a flow switch, two floats, a cyclic timer and an SMS alarm; one inverter for two clusters (about 5.8 hours). Plant and clear each cluster all at once. Keep the solution at or below 25–26 °C. Skip UV sterilisers. Use hydrogen peroxide only for cleaning: 37.5–75 mg/L in the solution stunted or killed lettuce.

Towers against other systems (lettuce)

System	Equipment ₹ per plant	Head weight	Survives a power cut for	Core ₹/kg	All-in ₹/kg
Towers, bought kit, pump each	₹167	90 g	0.5–1.0 h	₹78	₹126
Towers, your ₹3,000 body, 10-tower cluster	₹82	90 g	4–8 h	₹47	₹96
NFT flat channels	₹124	120 g	0.5–2 h	₹36	₹77
Raft beds (DWC), aerated	₹52	140 g	12–72 h	₹19	₹54
Kratky tanks, no pump (trial)	₹55	150 g	no power needed	₹16	₹51

Core = equipment over 5 years + power + labour per sellable kg. All-in adds seeds, nutrients, polyhouse and RO water. Dutch buckets (about ₹800 a plant), ebb-and-flow (about ₹1,200) and A-frames are poor fits. Power-cut hours for raft beds and most labour hours are estimates.

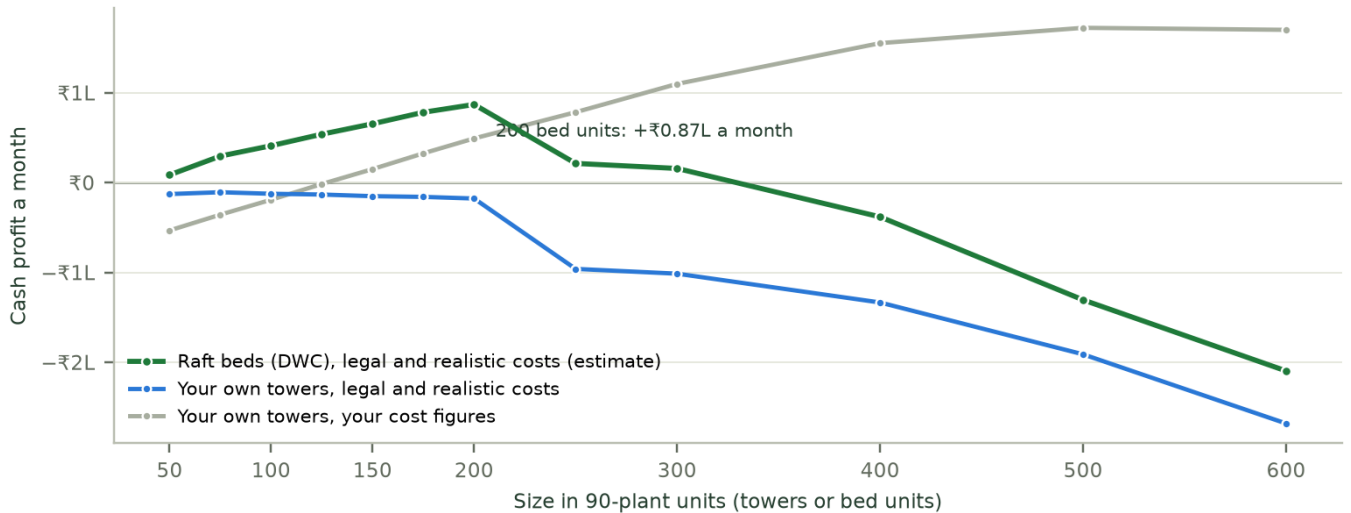
Recommendation. Build new capacity as aerated raft beds: 1.2 m wide, 25–30 cm deep, one solution per bed, one 100 W air pump per about 70 m² plus a spare, on an inverter. First run a pilot: 4–6 beds (about 2,000 plants) next to one 10-tower cluster (your 5 towers plus 5), for 2–3 crops. Measure head weight, labour, power, water, temperature and losses. Model pilot site cost: about ₹8.6 lakh plus ₹2.7 lakh for microgreen racks. Trial one Kratky bed as a no-power backup. The first fill needs about 250 L of RO water per m² of bed; beds also need mosquito control and a level floor.

Can 200 towers become 6,000 in 6 months? No.

What it needs	200 towers	600 towers	6,000 towers (indicative)	What you have
Money to build (bought kits)	₹50.9 lakh	₹1.49 crore	₹12.8 crore	₹20 lakh (₹4.4 lakh committed)
Covered area	380 m²	1,140 m²	11,400 m²	4,047 m² owned
Power connection	13 kW	31 kW	276 kW (high-voltage supply)	None yet; 15–90+ days
Workers	about 4	about 11	about 106	₹40,000 pays about 3
Tanker loads (12,000 L)	0.6 a day	1.9 a day	19 a day	Tankers only
Harvest a month	1.0 t	2.8 t	33 t	Buyers for about 1.7 t by month 12

Money breaks first. Above about 420 towers the harvest is more than buyers take at good prices. The model is not reliable above about 600–1,000 towers, so treat the 6,000 column as a rough guide.

Cash profit a month by size (base case)



Raft beds peak near 200 bed units, where their harvest meets month-12 demand. Own towers with legal costs stay below zero.

The fastest realistic path

When	What	Money	Go to the next step when
Oct-Dec 2026	Pilot: 4-6 raft beds + one 10-tower cluster (your 5 towers + 5). Microgreens up to 100 trays a week. Trade greens from partner farms. Test the water; apply for BESCOM LT-5 power, FSSAI and GST. No moulds.	About ₹11.3 lakh own cash; keep the other ₹8.7 lakh as reserve (it covers the ₹4.4 lakh of ₹40,000 payments)	2-3 crops measured; 10+ paying accounts; 60-95 kg a week sold at good prices
Jan-Mar 2027	Choose the system from the pilot. AIF loan application. Water: Cauvery or a permitted borewell, plus roof rainwater.	Fees	Loan approved; water secured
Jun-Sep 2027	Build 100 bed units, then up to 200, ready for the Oct-Nov season. Do not start new capacity in March-May.	₹25.7 lakh-₹42.9 lakh, mostly the loan	12 months of profit; 70%+ sold at good prices
2028 on	Add units only against signed demand. Moulds only if your own towers pass about 600.	From profit and loans	Signed demand for the extra output
Later	6,000 towers is a different business.	About ₹12-13 crore	Contracts in several cities; 3+ acres with water; high-voltage power; about 100 staff

Which crop earns the most (towers)

#	Crop	₹ per tower a month	₹ per tower per 40 days	Towers buyers take	Fit
1	Arugula (rocket)	₹1,246	₹1,640	27	good
2	Lollo Rosso lettuce	₹845	₹1,112	82	good
3	Mini romaine (Little Gem type)	₹715	₹941	73	good
4	Green oakleaf lettuce	₹679	₹894	85	good
5	Butterhead lettuce	₹645	₹849	56	fair
6	Batavia (summer crisp) lettuce	₹626	₹824	50	good
7	Italian (Genovese) basil	₹450	₹592	97	good
8	Pak choi (bok choy)	₹403	₹530	33	good
9	Swiss chard	₹366	₹482	11	good
10	Curly kale	₹357	₹470	96	fair
11	Thai basil	₹331	₹436	22	good
12	Flat-leaf parsley	₹148	₹195	24	good
13	English / baby spinach (Nov-Feb only)	₹65	₹85	37	poor
14	Mint	₹6	₹8	25	good
15	Palak (Indian spinach)	₹0	₹0	45	good
16	Coriander	-₹45	-₹59	44	fair
17	Strawberries (plant Oct-Nov, pick Dec-Apr)	-₹63	-₹83	10	fair

After seeds, media, nutrients, packing, delivery, commissions and unsold produce; before staff, power and water. Raft beds give about 1.5 times these kilograms per 90 plants (estimate). Season: Nov-Feb all lettuces, kale, pak choi, arugula; Mar-May Batavia and basil; Jun-Oct mildew-resistant lettuce, kale and chard.

What it costs to build

Item (base, GST included)	200 own towers	200 raft-bed units
Towers or beds, freight, assembly, spares	₹14,34,000	₹10,63,000
Polyhouse, fans, foggers, site work	₹5,97,900	₹15,87,600
Water: RO plant and tanks	₹3,58,419	₹3,58,419
Power: wiring, safety, backup, connection	₹4,44,794	₹2,27,561
Security: CCTV and fence	₹1,44,100	₹1,44,100
Post-harvest: cold storage and packing	₹1,02,000	₹1,02,000
People: welfare and caretaker room	₹1,84,000	₹1,84,000
Nursery, meters, registrations, marketing, first stock	₹2,35,800	₹2,35,800
Contingency (10%)	₹3,48,309	₹3,88,700
Moulds, design and tests (own towers, ring only)	₹12,40,000	-
Total	₹50,89,322	₹42,91,180

Every pain point, most serious first

Area	Pain point	Evidence
Selling	Buyers take far less than towers make	400 kg a week at good prices by month 12, against about 640 kg a week from 600 towers
Selling	Cheap, strong competitors	Hyperpure lettuce ₹140–216/kg, basil ₹150/kg; BigBasket 24 hydroponic lettuces at ₹47–85 a head
Selling	Floruvi list prices too high for volume	1.4–8.6 times Bengaluru shop prices
Selling	Late payment and failures	Restaurants pay in 15–60 days; Greenikk shut after ₹6 crore unpaid; Deep Rooted lost ₹52 crore
Money	Capital gap	200 raft-bed units ₹42.9 lakh; 200 own towers ₹50.9 lakh; you have ₹20 lakh
Money	Moulds before the first tower	₹12.4–26 lakh; own towers save money only above about 600
Money	Plastic price shock	PP ₹90.7 → ₹154.7/kg in 2026
Money	No subsidy in 3 months; land at risk	Approval before building; outside AIF, banks may ask to mortgage your land
Law	Wages below the legal minimum	₹10,000 guard and ₹15,000 7-day caretaker; minimum wage may rise 42% (court case)
Law	Design rights	Copying a registered design: claim up to ₹50,000 per design, or damages and an injunction
Growing	Small heads in towers	80–100 g; 31% lighter than flat systems; vendor claims are 1.5–3 times the research
Growing	Early losses, heat and root rot	20–35% at first; March–May lettuce –40%; one outbreak can kill a shared tank
Growing	Labour higher than planned	49 hours per 1,000 plants per 40 days (version 1 used 38)
Power	Pump stops kill tower crops	Roots dry in 30–60 minutes; inverters last 3–4 h; BESCOM shutdowns run 4–8 h
Water	Tanker water fragile and costly	₹2,000–2,850 a load in Feb–May; RO wastes half; hot months use 1.4–1.6×
Water	RO reject water has to go somewhere	About 3.8 m ³ a day at 200 towers
Equipment	Pumps and parts wear out	Pumps last about 2 years; film about 3 years; ‘no repairs’ holds for about 6 months
Equipment	Mould and part quality	Pocket undercuts, warping, strength of an 11-ring stack, UV life; basin needs a scarce 600–800 t press
Delivery	Delivery eats the margin at small volume	Restaurant drops cost ₹30–40/kg below 60 kg a day; home drops ₹127–400 in months 1–6
Time	No growing cash by December 2026	Towers ordered on 11 Oct give first lettuce about 2 Jan 2027
Scale	Model not valid at very large sizes	Treat anything above about 600–1,000 towers as a rough guide
You	Stress and rushed decisions	A day-and-night system, debt and family hopes. Grow in steps and keep a reserve

Research used

18 research agents read 724 unique sources on 29 September 2026. Two reviewers checked the model. Every link is in the appendix.

Topic	Key findings	Most-cited websites
Towers	Kit ₹9,500–13,500 plus GST; installed about ₹14,000; about 1.9 m ² per tower; pumps last about 2 years.	indiamart.com, dir.indiamart.com, greenwarrior.in, phooldaan.com
Polyhouse	Naturally ventilated polyhouse about ₹1,150/m ² ; shade-net houses let the monsoon rain in; film lasts about 3 years.	dir.indiamart.com, indiamart.com, horticulture.karnataka.gov.in, nhb.gov.in
Climate	Nov–Feb best; Mar–May lettuce –40%; Jun–Oct basil –30%; about 25% losses in the first 6 months.	en.wikipedia.org, uasbangalore.edu.in, pmc.ncbi.nlm.nih.gov, exportersindia.com
Power	LT-5 ‘Green House’ about ₹6.1 a unit all-in; LT-3 about ₹8.8; roots dry in 30–60 minutes without flow.	kerc.karnataka.gov.in, dir.indiamart.com, luminousindia.com, hrecs.co.in
Water	0.21 L per plant a day; salty groundwater, so RO (wastes about half); tankers about ₹92,000 a month at 600 towers.	deccanherald.com, dir.indiamart.com, thedailyjagran.com, cgwb.gov.in
Supplies	About ₹3.6 per lettuce plant per 40 days; own-mixed nutrients ₹170 per 1,000 L.	agriplexindia.com, tradeindia.com, flipkart.com, desikheti.com
Labour and security	Farm minimum wage ₹14,299 (₹20,350 in court); legal 12-h guard ₹61,000–75,000 by agency; CCTV about ₹83,000.	industrybuying.com, karmikasandana.karnataka.gov.in, apna.co, esic.gov.in
Lettuce	80–100 g a head in 90-site towers; Lollo Rosso and green oakleaf first; Batavia in Mar–May.	hyperpure.com, pmc.ncbi.nlm.nih.gov, en.wikipedia.org, indiamart.com
Herbs	30–50 g per plant per 40 days; restaurants pay ₹150/kg for basil; premium shops ₹750/kg in small amounts.	doi.org, hyperpure.com, trikaya.net, blr.onlyhydroponics.in
Other greens	Kale ₹144/kg, bok choy ₹136/kg at Hyperpure; palak, methi and amaranth lose money in towers.	bigbasket.com, johnnyseeds.com, hyperpure.com, kisandeals.com
Strawberries and microgreens	Microgreens: the only 30-day cash option; break-even about ₹700/kg. Strawberries weak.	tradeindia.com, doi.org, bigbasket.com, nayagreens.com
Prices	Hyperpure lettuce ₹140–216/kg; bulk ₹35–65/kg; BigBasket hydroponic heads ₹375–525/kg.	tradeindia.com, hyperpure.com, bigbasket.com, blr.onlyhydroponics.in
Demand and competition	New-seller demand 26 → 95 → 210 → 400 kg a week (months 1, 3, 6, 12).	exportersindia.com, entrackr.com, hyperpure.com, bigbasket.com
Packing and delivery	Delivery costs more than packing; EV three-wheeler ₹6–28/kg to restaurants; cold room ₹1.8–3.8 lakh.	tradeindia.com, borzodelivery.com, porter.in, trucks.cardekho.com
Export	Legal within 90 days; first shipment about 75 days away; only basil pilots with advance payment.	comtradeapi.un.org, apeda.gov.in, ppqs.gov.in, main.ecgc.in
Finance, law and risk	No subsidy pays in 3 months; AIF about 6%; NHB 35% and more than 4,000 m ² .	horticulture.karnataka.gov.in, agriinfra.dac.gov.in, sbi.bank.in, nhb.gov.in
Moulding and manufacturing	₹3,000 body realistic above about 2,200 towers; moulds ₹12.4–26 lakh; PP ₹154.7/kg; DIY PVC cheapest to 600 towers.	indiamart.com, dir.indiamart.com, made-in-china.com, plastemart.com
Pump clusters and systems	10 towers per 1,000 L tank; raft beds (DWC) cheapest per kg, 140 g heads, 12–72 h power-cut tolerance (estimate).	indiamart.com, dir.indiamart.com, agrotonomy.com, amazon.in

Key documents

Key document	Link
BESCOM tariffs FY2025-26 to FY2027-28 (KERC order, 27 March 2025)	kerc.karnataka.gov.in/uploads/85821743074692.pdf
Karnataka minimum wages 2026-27 (Labour Department)	karmikasandana.karnataka.gov.in/64/minimum-wages-rates-for-the-year-...
MIDH Operational Guidelines 2025 (subsidy norms)	nhb.gov.in/writereaddata/082825102800MIDH%20Guideline%202025.pdf
Agriculture Infrastructure Fund guidelines (September 2024)	agriinfra.dac.gov.in/Content/DocAttachment/FINALSchemeGuidelinesAIF.p...
IOCL polymer price list, 16 September 2026 (PP price)	plastemart.com/Upload/pricelist/16-09-2026-IOCL-Price-List.pdf
IP India design search	search.ipindia.gov.in/designsearch
IP India patent search	iprsearch.ipindia.gov.in/PublicSearch
Touliatos et al. 2016: tower against flat lettuce yield	pmc.ncbi.nlm.nih.gov/articles/PMC5001193/
University of Kentucky: hydroponic lettuce crop profile	ccd.uky.edu/sites/default/files/2024-11/ccd-cp-063_hydroponic-lettuce...
UAS Bangalore: hydroponic firms study (power cost per kg)	uasbangalore.edu.in/wp-content/uploads/2024/12/3.pdf
CGWB groundwater report, Bangalore Urban	cgwb.gov.in/sites/default/files/2022-10/bangalore_urban-2012.pdf
Hyperpure Bengaluru exotic vegetables (29 Sep 2026)	hyperpure.com/ind/bengaluru/exotic-vegetable
BigBasket lettuce listing (29 Sep 2026)	bigbasket.com/ss/lettuce/
Entrackr: Deep Rooted to shut operations	entrackr.com/exclusive/exclusive-accel-backed-deep-rooted-to-shut-ope...
Wikipedia: Bowery Farming (also AeroFarms, AppHarvest)	en.wikipedia.org/wiki/Bowery_Farming

Appendix: every source used

Towers

indiamart.com/proddetail/aeroponic-vertical-farming-with-tower-farm-2853449219612.html
indiamart.com/proddetail/aeroponics-grow-tower-26866084888.html
indiamart.com/proddetail/vertical-growing-tower-aeroponics-system-22695404512.html
indiamart.com/proddetail/aeroponics-grow-tower-25619349088.html
aajjo.com/product/plastic-hydroponic-innovative-vertical-aeroponic-tower-25-mm-sizedimension-15-meter-in-thane-aadya-hydro-solution
dir.indiamart.com/impcat/aeroponics-system.html
indiamart.com/proddetail/hydroponic-innovative-vertical-aeroponics-tower-compl
ete-set-up-21305317230.html
indiamart.com/proddetail/100-planter-vertical-hydroponic-aeroponics-tower-syste
m-made-by-vegtubes-2853243135130.html
indiamart.com/proddetail/hydroponic-vertical-tower-2850390934930.html
indiamart.com/proddetail/vertical-hydroponic-tower-2852006637288.html
vismayagroup.in/vertical-tower-vegetable-growing-hydroponics-unit.html
greenwarrior.in/shop
greenwarrior.in/product-page/nova-120-planter
greenwarrior.in/product-page/copy-of-nova-120-planter
phooldaan.com/nova-120
phooldaan.com/aeroponic-tower-wholesale
phooldaan.com/post/aeroponic-tower-price-in-india-full-cost-breakdown-monthly-r
unning-costs-roi-2026
hydrilla.in/hydroponics-190-plants-pyramid-stand.html/
indiamart.com/proddetail/hydroponic-nft-system-15326920255.html
indiamart.com/proddetail/90-plants-hydroponics-system-8904793773.html
agrinovix.en.made-in-china.com/product/nGzrumHUWRYM/China-Hydroponic-Verti
cal-Garden-Tower-for-Greenhouse-Food-Production.html (China ABS food-grade
UV towers, 20-36 holes, US\$80-110 FOB: not cheaper per site than Indi)
agrotonomy.com/what-is-the-average-cost-of-a-tower-farm/
agrotonomy.com/space-saving-of-a-vertical-farm-using-aeroponic-towers/
towerfarms.com/pages/faq
zipgrow.com/zipgrow-tower-spacing/
nhb.gov.in/pdf/Cost%20Norms.pdf
agropotli.com/polyhouse-subsidy-in-karnataka/
dir.indiamart.com/bengaluru/net-house.html
bartonbreeze.com/post/how-much-does-it-cost-to-set-up-a-one-acre-hydroponic-f
arm
static.pib.gov.in/WriteReadData/specificdocs/documents/2025/sep/doc202598630
601.pdf
essjayenterprises.com/sintex-water-tank-price-list-2026/
industrybuying.com/openwell-submersible-pump-kirloskar-PUM.OPE.623866011
indiamart.com/proddetail/digital-cyclic-timer-hydroponic-22172912633.html
risehydroponics.in/product/net-pots-2-inch/
indiamart.com/proddetail/net-cup-for-hydroponic-21305285388.html
indiamart.com/proddetail/hydroponic-net-cup-27045593130.html
indiamart.com/proddetail/98-cells-seedling-tray-or-seed-germination-tray-172371
53291.html
indiamart.com/proddetail/combo-ph-ec-tds-meter-pen-type-hanna-hi98129-28553
51213291.html
industrybuying.com/search/?q=ph%20meter%20pen
industrybuying.com/search/?q=ph%20buffer%20solution
indiamart.com/proddetail/500-lph-ro-plant-10581139891.html
romegamart.com/blog-details/1000-litre-ro-plant-price-in-india-5/
ijert.org/an-assessment-of-groundwater-quality-index-in-bommasandra-bengaluru
-city-karnataka-state-india
deccanchronicle.com/nation/current-affairs/070818/bengaluru-is-this-hard-tds-fille
d-water-safe.html
palmetto-industries.com/polypropylene-uv-resistance/
astralpipes.com/blogs/why-buy-lead-free-pipes/
dir.indiamart.com/impcat/upvc-pipes.html
citygreens.ai/blogs/post/vertical-farms-fail-india
aquahyd.in/buy-back-scheme/
ca.towergarden.com/products/pump (from search summary only: 5-year pump
warranty)
Not available: amazon.in (503), hannainst.in (403), inhydro.in (504), IndiaMART
meter category pages (429 rate limit), Sasyaved ₹5,000 tower listing (404,
excluded). The web-search limit was reached b

Polyhouse

nhb.gov.in/writereaddata/082825102800MIDH%20Guideline%202025.pdf (MIDH
Operational Guidelines 2025, dated 31.12.2024, Annexure V)
shm.tg.nic.in/Download/MIDH%20-%20Implementation%20Guidelines%20-%2020
25-26.pdf (Telangana MIDH 2025-26: net-house BOQ with 18% GST, NVPH spec,
process, insurance, 60-90 d)
horticulture.tg.nic.in/polyhouse/Downloads/Technical%20Specification.pdf
(warranty: 5 years structure, 3 years film)
horticulturedir.karnataka.gov.in/uploads/media_to_upload1758706503.pdf
(Karnataka NHM 2025-26 weed mat Rs 50/m2)
horticulturedir.karnataka.gov.in/uploads/media_to_upload1758695349.pdf
(Karnataka NHM 2025-26 unit costs, land preparation)
horticulturedir.karnataka.gov.in/info-2/Facilities+available+to+farmers/kn
(Karnataka portal, older polyhouse/shade-net subsidy figures)
horticulturedir.karnataka.gov.in/info-2/Central+Sector+Schemes/en (Karnataka
NHM component list)
hkkp.org.in/Other_Files/Details%20of%20scheme%20of%20Horticulture%20Dept
%20eng.pdf (Haryana 2022 state norms: cladding replacement, poly net house)
dir.indiamart.com/bengaluru/polyhouse.html
indiamart.com/proddetail/naturally-ventilated-polyhouse-14001460648.html
(Malur NVPH Rs 725/m2)
indiamart.com/proddetail/fan-pad-polyhouse-14806232948.html (fan-pad Rs
2,000/m2, listing title)
indiamart.com/proddetail/fan-pad-polyhouse-22831470297.html (fan-pad Rs
1,600/m2, listing title)
indiamart.com/proddetail/uv-stabilized-film-200micron-15264377248.html
indiamart.com/proddetail/anti-insect-net-40-mesh-18341868312.html
indiamart.com/proddetail/green-house-shade-net-2342934891.html
dir.indiamart.com/impcat/weed-control-mat.html

dir.indiamart.com/bengaluru/chain-link-fencing.html
dir.indiamart.com/bengaluru/barbed-wire-fencing.html
dir.indiamart.com/bengaluru/fencing-services.html
farmatma.in/polyhouse-types/
agropotli.com/polyhouse-farming-cost-per-acre-in-india/
agrifarming.in/hydroponic-farming-in-bangalore-how-to-start-setup-cost-companie
s-and-suppliers
agriplast.co.in/blog/polyhouse-subsidy-in-india
inhydro.in/product/naturally-ventilated-polyhouse/
agrodome.in/how-to-choose-between-polyhouse-vs-shade-net-house-for-your-far
m-criteria-climate-cost-and-crops/
en.wikipedia.org/wiki/Bangalore (climate table, IMD 1991-2020)
credlix.com/hsn-code/9406 (GST 18% on HSN 9406 greenhouses/prefabricated
buildings)

Climate

en.wikipedia.org/wiki/Template:Bengaluru_Weather_box (IMD 1991-2020 normals;
NOAA 1971-1990 sunshine)
power.larc.nasa.gov/api/temporal/climatology/point?parameters=ALLSKY_SFC_SW
_DWN (ALLSKY_SFC_PAR_TOT,T2M,T2M_MAX,T2M_MIN,RH2M,PRECTOTCORR,T2M
WET,WS2M&community=AG&longitude)
uasbangalore.edu.in/wp-content/uploads/2026/05/April-2026.E.pdf
uasbangalore.edu.in/wp-content/uploads/2026/06/May-2026.E.pdf
uasbangalore.edu.in/wp-content/uploads/2026/01/December-2025.pdf
uasbangalore.edu.in/wp-content/uploads/2024/11/october-e-2022.pdf
uasbangalore.edu.in/en/agrometeorology-gkvk-bengaluru/
weatherspark.com/y/108998/Average-Weather-in-Bengaluru-Karnataka-India-Year
-Round
mausamjournal.imd.gov.in/index.php/MAUSAM/article/view/1657
cpb-us-e1.wpmucdn.com/blogs.cornell.edu/dist/8/8824/files/2019/06/Cornell-CEA-
Lettuce-Handbook-.pdf
e-gro.org/pdf/E301.pdf
University of Kentucky CCD, Hydroponic Lettuce crop profile CCD-CP-63 (Kaiser &
Ernst), ccd.uky.edu
pmc.ncbi.nlm.nih.gov/articles/PMC5001193/
pmc.ncbi.nlm.nih.gov/articles/PMC5020074/
doi.org/10.3390/agronomy15122768 (abstract)
frontiersin.org/journals/plant-science/articles/10.3389/fpls.2024.1340641/full
pmc.ncbi.nlm.nih.gov/articles/PMC4865134/
pmc.ncbi.nlm.nih.gov/articles/PMC4439122/
ipm.ucanr.edu/agriculture/lettuce/downy-mildew/
aces.edu/blog/topics/crop-production/greenhouse-lettuce-production/
en.wikipedia.org/wiki/Lettuce
en.wikipedia.org/wiki/Bok_choy
en.wikipedia.org/wiki/Basil
en.wikipedia.org/wiki/Kale
hgic.clemson.edu/factsheet/spinach/
agriplast.co.in/blog/strawberry-farming-india-guide
sheelbiotech.com/strawberry-cultivation-in-india-cost-climate-complete-farming-g
uide/
jpds.co.in/performance-of-strawberry-cultivars-for-growth-yield-and-quality-unde
r-naturally-ventilated-polyhouse-conditions/
MIDH Operational Guidelines 2025 (dated 31.12.2024), Govt of India: protected
cultivation cost norms and add-on components
NHB/MIDH cost norms and pattern of assistance (XII plan): fan-pad Rs 1,400/m2,
naturally ventilated polyhouse Rs 844/m2, shade net Rs 710/m2
Telangana Horticulture technical specifications for polyhouses and shade net
houses (MIDH 2025-26): foggers, insect nets, thermal screen Rs 90/m2
BESCOM Electricity Tariff 2026-2028 (KERC order 27-03-2025),
bescom.karnataka.gov.in
dir.indiamart.com/impcat/water-chillers.html
indiamart.com/proddetail/1-ton-water-chiller-plant-20072616133.html
dir.indiamart.com/impcat/hydrogen-peroxide.html
exportersindia.com/indian-suppliers/shade-net.htm
exportersindia.com/indian-suppliers/air-circulator-fan.htm
exportersindia.com/indian-suppliers/uv-sterilizer.htm
exportersindia.com/indian-suppliers/greenhouse-fan.htm
water.usgs.gov/water-resources/software/DOTABLES/ (Benson & Krause oxygen
solubility method)
/Users/syedabddulmuqeeth/Developer/SandBox/floruvi.farm/docs/pricing-benchma
rks.csv
Working calculations: /private/tmp/claude-501-/Users-syedabddulmuqeeth-Develop
er-SandBox-floruvi-farm/c33029eb-8c54-4088-b935-27d0b3ae1386/scratchpad/a
gronomist/ (calc.py, cooling.py, lightbudget.py, n

Power

kerc.karnataka.gov.in/uploads/85821743074692.pdf (KERC: Tariff Charges
Approved for FY2025-26 to FY2027-28, order 27.03.2025)
kerc.karnataka.gov.in/uploads/97271743074381.pdf (KERC press note, Tariff
Order 27.03.2025: P&G surcharge 36 paise, rural rebate 20 paise, D)
kerc.karnataka.gov.in/143/tariff-order-2025/en (KERC Tariff Order 2025 document
index)
kerc.karnataka.gov.in/uploads/98041743075668.pdf (KERC order 27.03.2025
amending the P&G surcharge order of 18.03.2025)
hrecs.co.in/app/documents/TariffRates2026-27-28.pdf (KERC Electricity Tariff
2026 schedule: category applicability, LT supply up to 150 kW, IP-)
ESCOM tariff booklet 'Electricity Tariff 2026 to 2028' (KERC order 27.03.2025);
local copy: /private/tmp/claude-501-/Users-syedabddulmuqeeth-Developer-SandBo
x-floruvi-farm/c33029eb-8c54-4088-b935-27d0b
bescom.karnataka.gov.in/124/hours-of-supply-%28hos%29-to-all-category-of-cons
umers/kn (BESCOM hours of supply: 7 h 3-phase on agricultural feeders, 24x7 for
non-agricultural)
kerc.karnataka.gov.in/storage/pdf-files/Regulations/KERC%20(Security%20Deposi
t (%20(Second%20Amendment)%20Regulations,%202020_.pdf (KERC
CoS/RoE/SD amendments 11.12.2020:)
kerc.karnataka.gov.in/storage/repo/KERC(Security%20Deposit
(Regulations)2007DiscussionPaper.pdf (KERC Security Deposit Regulations 2007
draft: 75-day d)
pib.gov.in/PressReleaseIframePage.aspx?PRID=2008289 (MoP Electricity (Rights
of Consumers) Amendment Rules 2024: connection timelines)
etvbharat.com/en/state/bescom-cescom-cosumers-to-pay-more-bills-as-kerc-orde
rs-recovery-of-revenue-deficit-enn26041804681 (FY25 true-up basis)

powerpeakdigest.com/kerc-concludes-fy25-apr-cycle-with-mixed-true-up-outcome-s-across-all-licensees/ (APR order 17.04.2026, 56 paise)
newsable.asianetnews.com/karnataka-news/kerc-issues-true-up-tariff-order-besco-m-consumers-to-pay-56-paise-more-per-unit-articleshow-03ssvtv
starofmysore.com/power-bills-up-again-govt-slaps-pension-cess/amp/ (P&G surcharge 36/35/34 paise)
solarquarter.com/2026/04/02/kerc-extends-discounted-energy-rate-scheme-till-fy-28-eases-open-access-rules-for-consumers-in-karnataka/ (DERs order 31.03.2026)
billunits.in/state-calculators/bescom-bill-calculator/ (FPPCA range, bill components)
goodreturns.in/diesel-price-in-bangalore.html (diesel Rs 99.56/L, 29 Sep 2026)
bankbazaar.com/fuel/diesel-price-bangalore.html (diesel Rs 98.8/L, 29 Sep 2026)
dir.indiamart.com/impicat/mahindra-diesel-generator.html (DG listing prices)
dir.indiamart.com/impicat/kirloskar-diesel-generator.html (DG listing prices)
beeindia.gov.in/sites/default/files/3Ch9.pdf (BEE DG set guide: sizing, altitude derating, fuel use L/kWh)
luminousindia.com/high-capacity-inverters/all-products?eshop=true (inverter prices)
luminousindia.com/batteries/tubular-battery?eshop=true (150 Ah tubular battery prices)
lizahomes.in/blog/bescom-connection-smart-meter-guide-bengaluru-2026/ (smart meters mandatory for new connections; secondary source)
thediscombill.com/guides/bescom-new-connection-online/ (connection process and timelines; secondary source)
Cornell University CEA Program (2013), Hydroponic Lettuce Handbook; local copy scratchpad/cornell_lettuce.pdf (pump sizes)
Load and bill model: /private/tmp/claude-501/-Users-syedabdulmuqeeth-Develope-r-SandBox-floruvi-farm/c33029be-8c54-4088-b935-27d0b3ae1386/scratchpad/power/calc2.py

Water

thenewsminute.com/karnataka/bengaluru-district-administration-introduces-fixed-rates-for-water-tankers
thedailyjagran.com/india/bengaluru-water-crisis-bangalore-water-tanker-prices-in-crease-amid-emptying-borewells-check-rates-here-10226218
thedailyjagran.com/india/karnataka-govt-launches-first-gps-enabled-water-tankers-in-bengaluru-know-price-details-10236332
newsfirstprime.com/bengaluru/cauvery-pipes-state-tankers-loosen-bengalurus-tanker-mafia-grip-11836253
deccanherald.com/india/karnataka/bengaluru/bengaluru-water-crisis-scarcity-wor-sens-tanker-prices-surge-up-to-rs-2850-2913143
pressreader.com/india/the-hindu-international-9BN2/20240314/281676849885283
deccanherald.com/india/karnataka/bengaluru/karnataka-government-to-takeover-private-water-tankers-in-bengaluru-amid-severe-water-shortage-d...
deccanherald.com/india/karnataka/bengaluru/6-900-of-13-900-city-borewells-are-dry-says-dks-2932192
deccanherald.com/india/karnataka/bengaluru/bengaluru-80-of-borewells-fail-due-to-unscentific-drilling-says-hydrogeologist-3447377
deccanherald.com/india/karnataka/bengaluru/105-borewells-90-success-rate-bws-sb-s-secret-geologists-2979652
citizenmatters.in/borewell-digging-in-bengaluru-faqs-on-groundwater-extraction/biometrust.org/decoding-groundwater-regulations-for-borewells-in-bangalore/cgwb.gov.in/sites/default/files/2022-10/bangalore_urban-2012.pdf
cgwb.gov.in/cgwbnpnm/public/uploads/documents/169951116865609387file.pdf
neptjournal.com/upload-images/NL-6-12-12 (B-1231-69-76..pdf)
hort.cornell.edu/greenhouse/crops/factsheets/hydroponic-recipes.pdf
edepot.wur.nl/403810
flairform.com/hints/water_quality/
scielo.br/j/rbeaa/a/GVp5GHQJn56kBGdqvQpXbjg/?lang=en
zipgrow.com/water-conservation-in-zipgrow-systems/
agronomy.com/how-much-water-does-a-tower-garden-use/
indiamart.com/green-nova-technologies/hydroponic-system.html
greenwarrior.in/product-page/lotus-60-aeroponics-amp-hydroponics-vertical-garden-tower
dir.indiamart.com/impicat/reverse-osmosis-plant.html
dir.indiamart.com/impicat/ro-membranes.html
dir.indiamart.com/impicat/sintex-water-tanks.html
dir.indiamart.com/impicat/hdpe-pond-liner.html
bescom.karnataka.gov.in/uploads/media_to_upload1775021662.pdf
studiomatrix.org/guides/borewell-plumbing-cost-india
krishnaborewell.com/blog/borewell-cost/
MIDH Operational Guidelines 2025 (MoA&FW, dated 31.12.2024): water harvesting structures Rs 80/m3; hydroponics add-on Rs 350/m2 at 50% up to 1,000 m2

Supplies

desikheti.com/collections/buy-lettuce-seeds-online-at-best-price/products.json
desikheti.com/collections/rijk-zwaan-lettuce-seeds-buy-online/products.json
desikheti.com/collections/enza-zaden/products.json
bighaat.com/collections/rijk-zwaan
agrosiaa.com/products/detail/locarno-rz-f1-hybrid-85-81-leaf-lettuce-5000-pills
farmersstop.com/collections/lettuce-seeds-online-india
greeneryretail.com/product-page/dunand-by-rijk-zwaan-rz-crystal-lettuce-seeds-1000-pills
gardenguru.in/collections/premium-seeds-for-hydroponics-and-greenhouse-production
indiamart.com/proddetail/red-lettuce-shirane-sky-pelleted-johnny-seeds-1000-see-ds-2849425781448.html
dir.indiamart.com/bengaluru/lettuce-seeds.html
shop.indamseeds.com/products/lettuce-lollo-rosso
hpsopca.hp.gov.in/wp-content/uploads/2023/07/Indian_minimum_seed_certificati-on_standards.pdf
farmersstop.com/collections/basil-seeds-aromatic-medicinal-herb-varieties
flipkart.com/search?q=thai%20basil%20seeds
desikheti.com/collections/buy-kale-seeds-online-at-best-price/products.json
farmersstop.com/collections/buy-exotic-vegetable-seeds-online-in-india
farmersstop.com/collections/pakchoi-bok-choy-seeds-exotic-online-india
desikheti.com/collections/buy-spinach-seeds-online-at-best-price/products.json
indiamart.com/proddetail/swiss-chard-super-green-op-22430466348.html
bighaat.com/collections/swiss-chard
indiamart.com/proddetail/rocket-salad-seeds-21050948788.html

farmersstop.com/collections/coriander-seeds-online-in-india
thebetterindia.com/farming/mahabaleshwar-strawberry-farmers-35000-tonnes-success-story-11125671
dir.indiamart.com/impicat/strawberry-plants.html
tradeindia.com/products/tissue-culture-strawberry-plant-4511056.html
commodityonline.com/mandiprices/mint-pudina (search snippet only; page returned 403)
flipkart.com/search?q=hydroponic%20net%20cups%202%20inch
tradeindia.com/manufacturers/net-pots.html
flipkart.com/search?q=jiffy%20coco%20coir%20pellets%20seed%20starter
flipkart.com/search?q=rockwool%20starter%20cubes
flipkart.com/search?q=hydroponic%20grow%20sponge%20seed%20starter
nurserylive.com/search/suggest.json?q=hydroponic
trustbasket.com/search/suggest.json?q=hydroponic
organicbazar.net/search/suggest.json?q=hydroponic
agripalexindia.com/products/multiplex-nitrocal-calcium-nitrate-powder-1kg
agripalexindia.com/products/anshul-dual-1-kg
agripalexindia.com/products/multiplex-multi-pk-0-52-34
agripalexindia.com/products/katyayani_npk_00_52_34_fertilizer
agripalexindia.com/products/anshul-magnesium-magnesium-sulfate-9-50-1-kg
agripalexindia.com/products/multiplex-prokissan-chelated-multi-micronutrients
agripalexindia.com/products/multiplex-ruby-ferrous-iron-edta-12-0
agripalexindia.com/products/multiplex-mutical-g-calcium-chloride-1-kg
agripalexindia.com/products/margo-econeem-plus-10-000-ppm
agripalexindia.com/products/anshul-maxinemor-azadiractin-0-15-ec
agripalexindia.com/products/margo-delfin-bt
agripalexindia.com/products/multiplex-baba-beauveria-bassiana-powder1-kg
agripalexindia.com/products/multiplex-varsha-powder1-kg
agripalexindia.com/products/multiplex-attract-yellow-blue-sticky-traps-10-pcs
gardenguru.in/products.json (NPK 4-18-38 1 kg Rs 599)
dir.indiamart.com/impicat/calcium-nitrate.html
tradeindia.com/manufacturers/calcium-nitrate.html
tradeindia.com/manufacturers/potassium-nitrate.html
tradeindia.com/manufacturers/monopotassium-phosphate.html
tradeindia.com/manufacturers/magnesium-sulphate.html
tradeindia.com/manufacturers/ferrous-edta.html
tradeindia.com/manufacturers/phosphoric-acid.html
tradeindia.com/manufacturers/nitric-acid.html
tradeindia.com/manufacturers/potassium-hydroxide.html
flipkart.com/search?q=hydroponic%20nutrients%20a%20b
flipkart.com/search?q=ph%20down%20hydroponics
bighaat.com/collections/sticky-traps
flipkart.com/search?q=yellow%20sticky%20trap
dir.indiamart.com/impicat/sodium-hypochlorite.html
pmc.ncbi.nlm.nih.gov/articles/PMC4483736/ (Barbosa et al. 2015, IJERPH 12:6879)

Labour and security

karmikaspandana.karnataka.gov.in/64/minimum-wages-rates-for-the-year-2026-27/en
karmikaspandana.karnataka.gov.in/uploads/media_to_upload1774347727.pdf (Agriculture MW & VDA 2026-27)
karmikaspandana.karnataka.gov.in/uploads/media_to_upload1744032984.pdf (Agriculture MW & VDA 2025-26)
karmikaspandana.karnataka.gov.in/uploads/media_to_upload1774351875.pdf (Security Agency MW 2026-27)
karmikaspandana.karnataka.gov.in/uploads/media_to_upload1774350604.pdf (Food processing/packing MW 2026-27)
teamleaseregtech.com/updates/article/56206/karnataka-govt-issued-the-consolidated-revision-of-minimum-wages-for-8/
ascent-hr.com/notification/minimum-wages-basic-revision-karnataka-220526/
hrenalay.com/compliance/minimum-wages/karnataka
labourlawhelp.com/karnataka-minimum-wages/
saral.pro/blogs/minimum-wages-in-karnataka/
deccanherald.com/india/karnataka/karnataka-govt-hikes-minimum-wages-rs-23300-must-for-unskilled-work-in-bengaluru-4013828 (headline only; body paywalled)
livelaw.in/high-court/karnataka-high-court/karnataka-high-court-minimum-wage-notification-challenge-interim-protection-employers-551044
mondaq.com/india/employee-benefits-compensation/1799698/decoding-karnatak-as-revised-minimum-wage-notification
esic.gov.in/coverage
esic.gov.in/contribution
epfo.gov.in/ (FAQ: 12% + 12%, Rs 15,000 threshold)
en.wikipedia.org/wiki/Code_on_Wages_(2019)
en.wikipedia.org/wiki/Occupational_Safety_Health_and_Working_Conditions_Code_(2020)
ccd.uky.edu/sites/default/files/2024-11/ccd-cp-063_hydroponic-lettuce.pdf
freightfarms.com/blog/labor-requirements-for-the-greenery
apna.co/jobs/security_guard-jobs-in-bengaluru
apna.co/jobs/agriculture-jobs-in-bengaluru
apna.co/jobs/packing-jobs-in-bengaluru
apna.co/jobs/driver-jobs-in-bengaluru
apna.co/jobs/gardener-jobs-in-bengaluru
workindia.in/farm-worker-jobs-in-bangalore/
industrybuying.com/search/?q=cp+plus+8+channel+dvr
industrybuying.com/search/?q=4mp+ip+camera
industrybuying.com/search/?q=tp-link+vigi
industrybuying.com/search/?q=4g+router
industrybuying.com/search/?q=poe+switch+8+port
industrybuying.com/search/?q=security+siren
industrybuying.com/search/?q=led+flood+light+50w
industrybuying.com/search/?q=motion+sensor+light
industrybuying.com/search/?q=ups+for+cctv
industrybuying.com/search/?q=cat6+cable+305
industrybuying.com/search/?q=first+aid+kit
industrybuying.com/search/?q=fire+extinguisher+abc+4+kg
industrybuying.com/search/?q=nitrile+gloves
industrybuying.com/search/?q=gumboot
industrybuying.com/search/?q=pvc+apron
industrybuying.com/search/?q=barbed+wire
primeabgb.com/?s=skyhawk&post_type=product
primeabgb.com/?s=wd+purple&post_type=product
exportersindia.com/search.php?srch_catg_ty=prod&term=portable+toilet+cabin

Blocked or unavailable: IndiaMART (HTTP 429), Amazon.in (503), AmbitionBox (403), stalwartgroup.com security cost article (404), EPFO contribution-rate page (404), TeamLease copy of the 22 May 2026 no

Lettuce

cigrjournal.org/index.php/Ejournal/article/view/8435 (CIGR J. 2024, Philippines tower romaine 60.5-78.6 g at 30 DAT; verified)
agronomyjournals.com/archives/2025/vol8issue7S/PartB/S-8-7-20-352.pdf (Jain et al. 2025, Dapoli tower, 1.9 kg from 36 plants; verified from PDF)
pmc.ncbi.nlm.nih.gov/articles/PMC10322904/ (ICAR-IARI iceberg vertical NFT, 3-week nursery, 45 DAT, 275.5-393 g; verified)
zipgrow.com/a-focus-on-lettuce/ (10-14 d nursery, 4-5 weeks in tower, 0.7-1 lb per ft; verified)
agronomy.com/crop-yield-of-a-tower-farm/ (vendor: 100-175 g and 150-200 g per port, 52 ports; verified)
heraldopenaccess.us/openaccess/evaluation-of-nutrient-solution-on-the-growth-and-yield-performance-of-organic-lettuce-grown-in-vertical-tow... (organic towers, 1-63 g per plant)
pmc.ncbi.nlm.nih.gov/articles/PMC8190844/ (optimum 15-20 C; above 30 C promotes bolting)
pmc.ncbi.nlm.nih.gov/articles/PMC10667003/ (root-zone temperature effects)
en.wikipedia.org/wiki/Arugula (heat changes flavour and stops leaf growth)
almanac.com/plant/arugula (arugula regrowth and heat bolting; weak source)
bigbasket.com/ss/lettuce/ (raw catalogue JSON read 29 Sep 2026; default city cookie id 1)
bigbasket.com/ss/arugula/ (read 29 Sep 2026)
hyperpure.com/ind/bengaluru/exotic-vegetable (warehouse WH-BLR7, read 29 Sep 2026)
hyperpure.com/ind/bengaluru/lettuce-rocket-argula-250-gm
hyperpure.com/in/lettuce-rocket-argula-100-gm
hyperpure.com/in/green-lollo-rosso-lettuce-hydroponic-150-200-gm
hyperpure.com/in/lettuce-iceberg-1-kg
blr.onlyhydroponics.in/products.json (OnlyHydroponics Bengaluru feed, read 29 Sep 2026)
trikaya.net/products.json (Trikaya feed, read 29 Sep 2026)
dir.indiamart.com/bengaluru/lettuce.html (Sep 2026)
dir.indiamart.com/bengaluru/lettuce-iceberg.html (Sep 2026)
indiamart.com/proddetail/hydroponic-vegetable-lettuce-26464753488.html
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/pricing-benchmarks.csv (Floruvi list prices)
Scratch calculations: /private/tmp/claude-501-/Users-syedabdulmuqeeth-Developer-SandBox-floruvi-farm/c33029eb-8c54-4088-b935-27d0b3ae1386/scratchpad/rev40.py and rev40_results.txt

Herbs

Floruvi repo retail benchmarks: /Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/pricing-benchmarks.csv
hyperpure.com/ind/bengaluru/leafy-vegetables-1 (Hyperpure Bengaluru leafy vegetables (coriander, mint, dill))
hyperpure.com/ind/bengaluru/thyme-100-gm (Hyperpure Bengaluru thyme 100 g)
hyperpure.com/ind/bengaluru/rosemary-100-gm (Hyperpure Bengaluru rosemary 100 g)
healthybuddha.in/fruits-vegetables/fresh-leafy-greens (Healthy Buddha Bengaluru leafy greens and exotics; ; https://healthybuddha.in/fruits-vege)
bigbasket.com/ss/herbs/ (BigBasket herbs and exotic vegetables listing (anonymous session, location not confirmed):)
krama.karnataka.gov.in/ (KRAMA, Karnataka agricultural marketing daily statewide price table)
zipgrow.com/quickcropsguide/ (ZipGrow Quick Crops Guide (2020, PDF)
doi.org/10.21273/HORTTECH.25.5.645 (Walters & Currey 2015, HortTechnology 25:645 (basil systems and cultivars; abstract via Cr)
doi.org/10.21273/HORTSCI13126-18 (Walters & Currey 2018, HortScience 53:1319 (basil EC × DLI; abstract via Crossref)
doi.org/10.3390/agronomy9070389 (Currey, Walters & Flax 2019, Agronomy 9:389 (cilantro, dill, parsley NFT × DLI; abstract v)
doi.org/10.1371/journal.pone.0248662 (Walters & Lopez 2021, PLOS ONE 16:e0248662 (dill and parsley vs DLI and temperature)
doi.org/10.21273/HORTSCI12785-17 (Dou et al. 2018, HortScience 53:496 (sweet basil DLI)
vegetables.cornell.edu/pest-management/disease-factsheets/basil-downy-mildew/ (Cornell Vegetables, basil downy mildew fact sheet)
doi.org/10.1094/PHYTO-02-15-0032-FI (Wyenandt et al. 2015, Phytopathology 105:885 (basil downy mildew review)
Team scratch notes (seed, media and nutrient prices; IMSCS germination minima; Bengaluru climate normals): scratchpad/notes.md and scratchpad/agronomist/notes.md
Calculation script for yields and revenue: /private/tmp/claude-501-/Users-syedabdulmuqeeth-Developer-SandBox-floruvi-farm/c33029eb-8c54-4088-b935-27d0b3ae1386/scratchpad/herbs/herb_calc.py
Blocked or unavailable during this study: WebSearch budget used up; IndiaMART (HTTP 429), MDPI and ASHS full texts (403), UC Davis postharvest (403), HOPCOMS (DNS failure).

Other greens

hyperpure.com/ind/bengaluru/search/kale
hyperpure.com/ind/bengaluru/indian-vegetables-1
hyperpure.com/ind/bengaluru/search/chinese%20cabbage
bigbasket.com/ss/kale/
bigbasket.com/ss/pak-choi/
bigbasket.com/ss/hydroponic/
bigbasket.com/ss/palak/
bigbasket.com/ss/spinach/
bigbasket.com/ss/methi/
bigbasket.com/ss/amaranthus/
bigbasket.com/ss/spring-onion/
bigbasket.com/ss/mustard-leaves/
bigbasket.com/ss/chinese-cabbage/
bigbasket.com/ss/swiss-chard/
blr.onlyhydroponics.in/collections/all
organicmandya.com/products.json
healthybuddha.in/mustard-greens
kisandeals.com/mandiprices/AMARANTHUS/KARNATAKA/ALL
kisandeals.com/mandiprices/SPINACH/KARNATAKA/ALL

kisandeals.com/mandiprices/METHI(LEAVES (/KARNATAKA/ALL)
kisandeals.com/mandiprices/ONION%20GREEN/KARNATAKA/ALL
kisandeals.com/mandiprices/CABBAGE/KARNATAKA/ALL
vegetablemarketprice.com/market/bangalore/today
agritech.tnau.ac.in/horticulture/horti_vegetables_amaranthus.html
agritech.tnau.ac.in/horticulture/horti_vegetables_minor%20vegetables.html
cea.cals.cornell.edu/files/2019/06/Cornell-CEA-baby-spinach-handbook.pdf
agronomy.com/choosing-regular-towers-over-high-density-towers-when-starting-a-tower-farm/
agronomy.com/what-does-not-grow-on-hydroponic-towers/
agronomy.com/growing-cabbages-on-aeroponic-towers/
pmc.ncbi.nlm.nih.gov/articles/PMC12992682/
pmc.ncbi.nlm.nih.gov/articles/PMC12425742/
johnnyseeds.com/vegetables/kale/
johnnyseeds.com/vegetables/swiss-chard/
johnnyseeds.com/vegetables/spinach/
johnnyseeds.com/vegetables/chinese-cabbage/
johnnyseeds.com/vegetables/onions/bunching-onions/evergreen-hardy-white-onion-seed-502.html
johnnyseeds.com/vegetables/greens/pac-choi-bok-choy/mei-qing-choi-f1-green-seed-509.html
desikheti.com/products.json
doi.org/10.15406/hij.2019.03.00122
Mattson N.S. & Peters C., 'A Recipe for Hydroponic Success' (Cornell) - local research copy, URL not verified
Model script: /private/tmp/claude-501-/Users-syedabdulmuqeeth-Developer-SandBox-floruvi-farm/c33029eb-8c54-4088-b935-27d0b3ae1386/scratchpad/greens/greens40.py

Strawberries and microgreens

doi.org/10.3389/fpls.2023.1220691
doi.org/10.3389/fpls.2023.1177844
doi.org/10.3389/fpls.2025.1469430
doi.org/10.1080/14620316.2024.2449026
doi.org/10.21273/horttech.26.4.466
doi.org/10.33545/2618060x.2024.v7.i9c.1481
doi.org/10.20546/ijcmas.2020.910.325
doi.org/10.20546/ijcmas.2020.909.030
doi.org/10.62029/jmaps.v4i4i1.singh
nhb.gov.in/model-project-reports/Horticulture%20Crops/Strawberry/Strawberry1.htm
nayagreens.com/microgreens-live/
nayagreens.com/wp-json/wc/store/v1/products
nutrifarms.com/wp-json/wc/store/v1/products
bigbasket.com/ss/microgreens/
bigbasket.com/ss/strawberry/
bigbasket.com/ss/dried-green-peas/
bigbasket.com/ss/mustard-seeds/
organicbazar.net/products.json
farmersstop.com/products.json
ugao.com/products.json
tradeindia.com/search.html?keyword=strawberry+plant
tradeindia.com/search.html?keyword=fresh+strawberry
tradeindia.com/search.html?keyword=tissue+culture+strawberry+plant
tradeindia.com/products/sunflower-microgreen-seeds-10006700.html
tradeindia.com/search.html?keyword=radish+seeds
tradeindia.com/products/fresh-onion-microgreens-10417176.html
tradeindia.com/search.html?keyword=microgreen+tray
tradeindia.com/search.html?keyword=slotted+angle+rack
tradeindia.com/search.html?keyword=led+tube+light+20w
tradeindia.com/search.html?keyword=cocopeat+block+5+kg
tradeindia.com/search.html?keyword=strawberry+punnet
tradeindia.com/search.html?keyword=clamshell+container
flipkart.com/search?q=microgreens+tray+10x20
bescom.karnataka.gov.in/storage/pdf-files/RA%20section/ElectricityTariff2025.pdf
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/15-live-trays-and-pricing.md
/private/tmp/claude-501-/Users-syedabdulmuqeeth-Developer-SandBox-floruvi-farm/c33029eb-8c54-4088-b935-27d0b3ae1386/scratchpad/sm/calc_sm.py

Prices

hyperpure.com/ind/bengaluru/lettuce-loloroso-red-250-gm
hyperpure.com/ind/bengaluru/lettuce-romain-250-gm
hyperpure.com/ind/bengaluru/lettuce-iceberg-200-500-gm-1-pc
bigbasket.com/faq/
naturesbasket.co.in/search?q=lettuce (plus basil, kale, pak choi, bok choy, parsley, mint, microgreens)
healthybuddha.in/index.php?route=product/search&search=lettuce (plus other crops)
organicmandya.com/products.json
tradeindia.com/manufacturers/lettuce.html
tradeindia.com/manufacturers/romaine-lettuce.html
tradeindia.com/manufacturers/celery.html
tradeindia.com/manufacturers/parsley-leaves.html
tradeindia.com/manufacturers/mint-leaves.html
tradeindia.com/manufacturers/coriander-leaves.html
tradeindia.com/search.html?keyword=thai+basil (plus fresh strawberry, kale, pak choi, arugula, swiss chard, baby spinach, microgreens)
api.agmarknet.gov.in/v1/prices-and-arrivals/date-wise/specific-commodity (Agmarknet 2.0 public read API; Karnataka, Aug-Sep 2026)
sell.amazon.in/sell-online/product-categories/grocery
techcrunch.com/2026/06/03/firstclub-doubles-valuation-to-255m-in-nine-months-on-quality-first-grocery-bet/
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/11-pricing-research.md
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/22-growth-channels-research.md (Blinkit, Zepto, Instamart and BigBasket onboarding; S33-S39)
Blocked or unavailable on 29 Sep 2026: blinkit.com (Cloudflare 403), zeptonow.com and swiggy.com/instamart (bot-check 202), dir.indiamart.com (429), hopcoms.kar.nic.in (DNS failure), namdharifresh.com
Working files: /private/tmp/claude-501-/Users-syedabdulmuqeeth-Developer-SandBox-floruvi-farm/c33029eb-8c54-4088-b935-27d0b3ae1386/scratchpad/pa/perkg.py, hp_crawl_out.txt, bb_ss_*.txt)

Demand and competition

Method note: the web-search allowance for this session (200 of 200 searches) was used up before this task began. All evidence comes from pages fetched directly on 29 Sep 2026. Restaurant counts, amoun

hyperpure.com/
bigbasket.com/ss/exotic-vegetables/
blr.onlyhydroponics.in/pages/about-us
trikaya.net/pages/about-us
simplyfresh.co.in
bartonbreeze.com
ninjacart.com
firstclub.co.in/
saladays.co
curefoods.in
exportersindia.com/bangalore/lettuce.htm
exportersindia.com/krishnagiri/lettuce.htm
exportersindia.com/ooty/lettuce.htm
exportersindia.com/nilgiris/lettuce.htm
exportersindia.com/chittoor/lettuce.htm
exportersindia.com/bangalore/exotic-vegetables.htm
entrackr.com/exclusive/exclusive-accel-backed-deep-rooted-to-shut-operations-8865073
web.archive.org/web/20240526001333/https://deeprooted.co/
entrackr.com/2023/03/accel-backed-clover-ventures-gmv-grows-over-3x-in-fy22/
entrackr.com/2024/04/otipy-posts-50-gmv-growth-in-fy24-losses-down-by-21/
entrackr.com/tag/waycool/
entrackr.com/news/ninjacart-raises-6-mn-after-45-years-claims-ebitda-profitability-ahead-of-ipo-12128023
entrackr.com/2024/09/agritech-greenikk-shuts-down-returns-partial-capital-investors/
thebetterindia.com/307986/ahmedabad-friends-offer-training-earn-crores-with-ris-e-hydroponics-startup/
thebetterindia.com/444372/mohit-nijhawan-greenu-microgreens-farming-organic-miniature-vegetables-training-farmers/
gourmetgarden.in/ (Shopify page 'Store unavailable', HTTP 402, on 29 Sep 2026; the site was live in Sep 2025)
web.archive.org/web/20220121192824/https://www.urbankisaan.com/ (by Jan 2022, Urbankisaan had changed from its own hydroponic farms to selling produce from)
en.wikipedia.org/wiki/Hosur
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/22-growth-channels-research.md
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/31-growth-tools-plan.md

Packing and delivery

borzodelivery.com/in/tariffs (Bengaluru region set through
https://borzodelivery.com/in/bangalore), fetched 29 September)
borzodelivery.com/in/for_small_business
borzodelivery.com/in/business-api/docs
porter.in/trucks/bangalore
porter.in/two-wheelers/bangalore
uber.com/in/en/item-delivery/
smartprix.com/bytes/ola-launches-parcel-service-across-india-challenging-swiggy-genie-for-intracity-deliveries/
trucks.cardekho.com/en/trucks/mahindra/treo-zor
trucks.cardekho.com/en/trucks/euler/hiload-ev
bikedekho.com/tvs/xl-100
bikedekho.com/honda/activa/price-in-bangalore
bankbazaar.com/fuel/petrol-price-bangalore.html
goodreturns.in/petrol-price-in-bangalore.html
apna.co/jobs/delivery-person-jobs-in-bengaluru
karmikaspadana.karnataka.gov.in/uploads/media_to_upload1774351947.pdf
tradeindia.com/products/750-ml-pet-hinged-box-c12100106.html
tradeindia.com/products/250-gm-oyster-mushroom-packaging-punnets-c12100318.html
tradeindia.com/products/120gm-pet-punnet-box-c6130097.html
tradeindia.com/products/salad-bowl-medium-c7427871.html
tradeindia.com/products/maxifresh-map-bags-with-microholes-c10644343.html
tradeindia.com/products/50x25mm-direct-thermal-barcode-label-c11728755.html
tradeindia.com/products/3-x-1-5-inch-printed-barcode-label-c11311419.html
tradeindia.com/products/tsc-tp7-244-pro-barcode-printer-c11311410.html
tradeindia.com/products/1976-3-heavy-duty-40-l-fruits-and-vegetables-crates-c11127860.html
tradeindia.com/products/supreme-fruits-and-vegetable-crate-c2917827.html
tradeindia.com/products/250-gram-ice-gel-pouch-pack-c7186788.html
tradeindia.com/products/disposable-cold-ice-packs-c1139788.html
tradeindia.com/products/ice-gel-pack-c11559849.html
tradeindia.com/products/8-ltrs-thermocool-medicine-boxes-c574913.html
tradeindia.com/products/normal-eps-thermocool-box-c7928749.html
tradeindia.com/products/thermocool-ice-box-c11443721.html
tradeindia.com/products/food-delivery-insulated-bags-9046860.html
tradeindia.com/products/home-delivery-bag-c12502285.html
tradeindia.com/products/pp-laminated-kraft-paper-bag-c260016.html
tradeindia.com/products/vegetable-storage-cold-room-c10427168.html
tradeindia.com/products/2-ton-vegetable-cold-storage-c11268797.html
tradeindia.com/products/vegetable-cold-rooms-c9632644.html
tradeindia.com/products/380-v-three-phase-fully-automatic-cold-room-c8261321.html
tradeindia.com/products/8-ton-cold-storage-rooms-c12610400.html
tradeindia.com/products/western-single-door-visi-cooler-c9988797.html
tradeindia.com/products/05-blue-star-vc325d-western-visi-cooler-c10645991.html
tradeindia.com/products/535-ltr-daikin-deep-freezer-c12074485.html
tradeindia.com/products/bluestar-310-0-l-cf330neww-frost-free-double-door-hard-top-chest-deep-freezer-c12726820.html
tradeindia.com/products/commercial-salad-spinner-c9525082.html
tradeindia.com/products/100-kg-digital-platform-weighing-scale-c13325085.html
tradeindia.com/products/table-top-scales-c8385865.html
indiamart.com/proddetail/10l-eps-ice-thermocool-box-2854729726362.html (through docs/28 R45; IndiaMART rate-limited on 29 September)
seller.hyperpure.com/ (Seller Hub app text: FAQ, bid terms, onboarding flow)
brands.zepto.co.in/ (portal meta description)

/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/22-growth-channels-research.md (Blinkit, Zepto, Instamart, BigBasket onboarding; checked 27 September 2026)
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/28-shipping-and-languages-research.md (shelf life, EPS box, same-city options)
foscos.fssai.gov.in/assets/docs/Revised_2ndApril2026KindofBusinessEligibility.pdf
fssai.gov.in/upload/advisories/2026/03/69b4054bb6cd6Order%20dated%2013thMarch2026_Revised%20Turnover%20threshold.pdf
admin.gs1india.org/uploads/GCP_Fees_e3ed978cac.pdf
gs1india.org/register-for-barcodes
gs1india.org/get-a-barcode
ipindia.gov.in/pages/trade-marks/learn/forms-and-official-fees
udyamregistration.gov.in/
mofpi.gov.in/sites/default/files/study_report_of_post_harvest_losses.pdf
midh.gov.in/ (MIDH Operational Guidelines 2025, dated 31.12.2024, section C ; the team's scratchpad has)
Calculation script: /private/tmp/claude-501-Users-syedabdulmuqeeth-Developer-SandBox-floruvi-farm/c33029eb-8c54-4088-b935-27d0b3ae1386/scratchpad/pkgc-0ld/logistics_calc.py
Note: the session's WebSearch quota was used up. All data comes from direct page fetches. These pages could not be read: IndiaMART (HTTP 429), Blinkit seller pages (Cloudflare 403), Swiggy (bot check)

Export

apeda.gov.in/sites/default/files/documents/2026-06/Registration_Procedure.pdf
apeda.gov.in/RMCMC
dgft.gov.in/CP/index.jsp?opt=e-rcmc
content.dgft.gov.in/Website/dgftprod/62923d1c-9384-4416-abab-7d9cf488c156/2K_Updated.pdf (via repo docs/22 S91)
ppqs.gov.in/faq/plant-quarantine
ppqs.gov.in/divisions/plant-quarantine/import-export-procedure
ppqs.gov.in/divisions/plant-quarantine/present-setup
pqms.cgg.gov.in/pqms/getExportFee
iihr.res.in/nabl-accredited-food-testing-lab
apeda.gov.in/sites/default/files/Financial%20Assistance%20Schemes/FAS_Guidelines_05102021.pdf
apeda.gov.in/exporter-procedures
apeda.gov.in/FreshFruitsAndVegetables
kappec.karnataka.gov.in/storage/pdf-files/Agri%20Export%20Cell%20Files/KAPPECAssistance.pdf
main.ecgc.in/small-exporters-policy-sep-2
main.ecgc.in/shipments-comprehensive-risks-policy-scr
rbi.org.in/Scripts/BS_ViewMasDirections.aspx?id=10395
comtradeapi.un.org/public/v1/preview/C/A/HS?reporterCode=699&period=2024&partnerCode=784 (682,634,414,512,48&cmdCode=070999,070511,070519,070700,070960&flowCode=X)
comtradeapi.un.org/public/v1/preview/C/A/HS?reporterCode=784&period=2023&cmdCode=070511 (070519,070999,121190&flowCode=M)
comtradeapi.un.org/public/v1/preview/C/A/HS?reporterCode=682&period=2024&cmdCode=070511 (070519,070999,121190&flowCode=M)
comtradeapi.un.org/public/v1/preview/C/A/HS?reporterCode=634&period=2024&cmdCode=070511 (070519,070999,121190&flowCode=M)
comtradeapi.un.org/public/v1/preview/C/A/HS?reporterCode=699&period=2023&partnerCode=784&cmdCode=070999 (070519,070511,121190,060311&flowCode=X)
comtradeapi.un.org/public/v1/preview/C/A/HS?reporterCode=784&period=2023&partnerCode=699&cmdCode=060311 (080450,081090,070700,070960&flowCode=M)
talabat.com/uae/talabat-mart/fruit-veg/herbs-leafy-greens
talabat.com/uae/talabat-mart/fruit-veg/fresh-vegetables
trade.gov/country-commercial-guides/united-arab-emirates-import-requirements-and-documentation
trade.gov/knowledge-product/united-arab-emirates-labelingmarking-requirements
ahircargonews.net/editorial/2026/02/emirates-skycargo-deploys-two-additional-freighters-to-india/
onslog.com/resources/shipping-cost-from-india-2026
open.er-api.com/v6/latest/USD
mocca.gov.ae/en/services/import-permit (blocked: Request Rejected)
zad.gov.ae (connection refused; not verified)
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/22-growth-channels-research.md (section 6; S102 Intertek Saudi CoC; S103 MOCCAE)
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/28-shipping-and-languages-research.md (A4: volume weight R54, CEPA CoO R58)
/Users/syedabdulmuqeeth/Developer/SandBox/floruvi.farm/docs/market-pricing/research/ae.json (UAE retail herbs, Sep 2026)
Model script: /private/tmp/claude-501-Users-syedabdulmuqeeth-Developer-SandBox-floruvi-farm/c33029eb-8c54-4088-b935-27d0b3ae1386/scratchpad/landed.py

Finance, law and risk

nhb.gov.in/writereaddata/MIDH_Guidelines.pdf (MIDH Operational Guidelines, April 2025)
nhb.gov.in/online_application_nhb_scheme_2020_21.aspx?&menu.Menu=30124 (NHB scheme guidelines page: Public Circular dated 21-08-2026; MoA&FW letter M-140115/5/2024)
agriinfra.dac.gov.in/Content/DocAttachment/FINALSchemeGuidelinesAIF.pdf (AIF Revised Scheme Guidelines, Sept 2024)
agriinfra.dac.gov.in/Home/FAQs
agriinfra.dac.gov.in/Documents/Publication/E66DE2F0EBB143E8A688299807207130.pdf (AIF newsletter Vol.3, June-July 2026)
sbi.bank.in/web/interest-rates/interest-rates/agricultural-segment (Agriculture Segment Interest Structure as on 15-09-2026)
sbi.bank.in/web/interest-rates/interest-rates/mclr
sbi.bank.in/web/interest-rates/external-benchmark-based-lending-rate
rbi.org.in/ (policy repo 5.25%, checked 29-09-2026)
mudra.org.in/Offerings
standupmitra.in/Home/SUISchemes
horticulture.karnataka.gov.in/36/central-sector-schemes/en
horticulture.karnataka.gov.in/uploads/media_to_upload1758700999.pdf (Karnataka MIDH 2025-26 post-harvest management guideline)
horticulture.karnataka.gov.in/uploads/media_to_upload1758699345.pdf (Karnataka MIDH 2025-26 individual farm pond)
horticulture.karnataka.gov.in/uploads/media_to_upload1758277470.pdf (Karnataka Horticulture budget 2025-26)

gstcouncil.gov.in/sites/default/files/2025-09/press_release_press_information_bur
eau_0.pdf (56th GST Council recommendations)
cbic-gst.gov.in/pdf/chapter-wise-rate-wise-gst-schedule-18.05.2017.pdf
prsindia.org/files/bills_acts/bills_parliament/2025/The_Income-tax_Bill_(2025.pdf
and https://prsindia.org/billtrack/the-income-tax-bill-2025)
FSSAI FoSCoS 'Kind of Business' eligibility criteria, updated 01-04-2026
(foscoss.fssai.gov.in)
indiafilings.com/gst-return-filing (https://www.indiafilings.com/itr-filing ;
https://www.indiafilings.com/business-plan ; htt)
kerc.karnataka.gov.in/uploads/96731743148968.pdf (KERC Combined ESCOMS
Tariff Order 2025) and https://kerc.karnataka.gov.in/143/tariff-order)
University of Kentucky CCD Crop Profile CCD-CP-63 'Hydroponic Lettuce'
(uky.edu/ccd)
Cornell CEA Hydroponic Lettuce Handbook (cornellcea.com)
Telangana MIDH Implementation Guidelines 2025-26 (polyhouse insurance and
instalment rules)
CGWB Aquifer Mapping and Management Plan, Bengaluru South
(SWR/RP/NQM/2022-23/57)
en.wikipedia.org/wiki/AeroFarms (https://en.wikipedia.org/wiki/AppHarvest ;
https://en.wikipedia.org/wiki/Bowery_Farming (c)
PMEGP (kvionline.gov.in) did not resolve on 29-09-2026, and the PMFME portal
(pmfme.mofpi.gov.in) needs JavaScript, so I could not check their figures again
today

Moulding and manufacturing

Earlier Floruvi research in this session (scratchpad research.json: kit prices, DIY
PVC tower, labour rates)
antienindustries.com/en/product/additive-masterbatch/uv-masterbatch
brainly.in/question/4224255 (rule of thumb, search summary only; page blocked)
clutch.co/in/agencies/product-design/bengaluru
consulting.teqzo.com/top-product-design-companies-in-india/
dir.indiamart.com/delhi/injection-molding-job-works.html (450 t job-work rate
from search-result text)
dir.indiamart.com/delhi/used-injection-molding-machines.html
dir.indiamart.com/impcat/crate-mould.html
dir.indiamart.com/impcat/milacron-injection-molding-machines.html
dir.indiamart.com/impcat/plastic-bucket-mould.html
dir.indiamart.com/impcat/plastic-injection-moulding-machine.html
dir.indiamart.com/impcat/plastic-testing-services.html
dir.indiamart.com/impcat/plastic-tub-mould.html
dir.indiamart.com/impcat/uv-stabilizer-masterbatch.html
dir.indiamart.com/impcat/vacuum-forming-moulds.html
dir.indiamart.com/rajkot/plastic-injection-mold.html
eximpe.com/hsncode-finder/39269099
gnhydroponic.en.made-in-china.com/product/UfJGqzPTqIAo/China-Commercial-Hy
droponic-Vertical-Tower-Planter-Farming-Grow-System.html
iamrapid.com/pricing/
in.indeed.com/career/injection-mold-operator/salaries/Bengaluru--Karnataka
indiankanoon.org/doc/1643916/
indiankanoon.org/doc/991805/
iprsearch.ipindia.gov.in/PublicSearch
jcadusa.com/vacuum-casting-cost-2025/ (search summary only)
jchtrade.com/notes/what-an-injection-mold-really-costs/
kennedyguan.en.made-in-china.com/product/GFmjDBIKaukt/China-Stackable-Verti
cal-Planter-Pot-Garden-Tower-Ideal-for-Growing-Strawberries-Herb...
kennedyguan.en.made-in-china.com/product/LdhjnVNOFukM/China-Commercial-G
rowing-Bucket-Stackable-Tower-Pot-Planter-for-Strawberry.html
patentfilingcost.com/patent-registration-cost-in-india/
patents.google.com/patent/US10136594
patents.google.com/patent/US10729080
precious3d.com/injection-molding-cost-in-india-2026-tooling-per-unit-hidden-costs
/
printform.com/why-your-injection-molding-scrap-rate-is-higher-than-quoted/
search.ipindia.gov.in/designsearch
unimarkslegal.com/prior-art-search-services/
aadyahydro.in/hydroponic-system.html
ctrludhiana.org/wp-content/uploads/2021/07/Machine-Hour-Rates.pdf
daobenmachinery.com/injection-molding-machine-price-list/
enoshinfra.com/industrial-sheds-factory-for-rent-bangalore/
exchangerates.org.uk/USD-INR-spot-exchange-rates-history-2026.html
eximguru.com/indian-customs-duty/84807100-moulds-for-rubber-or-plastics.aspx
gimbooks.com/blog/hsn-code-3926-other-plastics-materials-gst-rate/
indiacode.nic.in/bitstream/123456789/1917/1/A2000-16.pdf
indiamart.com/proddetail/20-litre-plastic-bucket-mould-hot-runner-24621075688.
html
indiamart.com/proddetail/40-tio2-white-masterbatch-15111498448.html
indiamart.com/proddetail/450-ton-injection-moulding-machine-19731668497.html
indiamart.com/proddetail/haitian-120-ton-servo-injection-molding-machine-28554
98089812.html
indiamart.com/proddetail/haitian-650-ton-injection-molding-machine-2059584013
0.html
indiamart.com/proddetail/hp-mjf-3d-printing-services-27218333855.html
indiamart.com/proddetail/hydroponic-net-pots-plastic-injection-mould-264632244
30.html
indiamart.com/proddetail/injection-molding-chiller-21722601773.html
indiamart.com/proddetail/plastic-ghamela-mould-25490467933.html
indiamart.com/proddetail/plastic-injection-mould-p-20-matrial-20195409855.html
indiamart.com/proddetail/pp-sheet-13627388197.html
indiamart.com/proddetail/rotomoulding-mould-for-underground-vertical-and-septi
c-water-tank-rotational-mould-20705392788.html
industrybuying.com/dustbins-and-waste-bins-nilkamal-CLE.WAS.54905763
induthermofomers.com/post/how-much-does-a-thermoforming-tool-cost-in-india
kerkeextruder.com/news/industry-news/comprehensive-guide-to-anti-uv-masterb
atch-and-weather-resistance-masterbatch-outdoor-durability-and-t...
made-in-china.com/products-search/hot-china-products/Bucket_Mould.html
made-in-china.com/products-search/hot-china-products/Hydroponic_Net_Pot.html
made-in-china.com/products-search/hot-china-products/Hydroponic_Tower.html
made-in-china.com/products-search/hot-china-products/Net_Pot_Mould.html
made-in-china.com/products-search/hot-china-products/Plant_Pot_Mould.html
made-in-china.com/products-search/hot-china-products/Planter_Mould.html
moldrite.in/blog/injection-molding-cost-india
ofbusiness.com/prices/polymers-packaging
plastemart.com/Upload/pricelist/01-01-2026-IOCL-Price-List.pdf

plastemart.com/Upload/pricelist/01-06-2026-IOCL-Price-List.pdf
plastemart.com/Upload/pricelist/16-09-2026-IOCL-Price-List.pdf
plastemart.com/polymer-price-movement
procurementresource.com/news-and-articles/india-plastic-manufacturing-margin-
pressure-pvc-pp-pe-prices
protolabs.com/resources/blog/aluminum-mold-tooling-for-injection-molding/
sciencedirect.com/science/article/pii/S092134492500607X
seaaircargosystems.com/blog-china-india-freight-rates-2026.html
tradeindia.com/products/70-ltr-plastic-tub-7400092.html
tradeindia.com/products/injection-moulding-machine-650-ton-9081929.html
truelancer.com/cad-freelancers (search summary only)
zy-mold.com/flower-pot-mould
zetarmold.com/injection-mold-price-list-2026/

Pump clusters and systems

30stades.com/farming/kerala-brothers-build-one-of-indias-largest-hydroponic-lett
uce-farms-wayanad-export-to-dubai-qatar-9378532
agrotonomy.com/pumps-used-in-a-tower-farm/
agrotonomy.com/tower-farm-automated-irrigation/
aquaponics.africa/aeration-in-aquaponics/
asuveg.com.au/app/data/technical-insights/docs/VG04012_complete.pdf
ccd.uky.edu/sites/default/files/2026-02/ccd-cp-63_hydrolettuce-updated_accessibl
e.pdf
cropping.com/blogs/knowledge-center/nft-production-lettuce
dir.indiamart.com/impcat/float-switches.html
dir.indiamart.com/impcat/paddle-flow-switch.html
dir.indiamart.com/impcat/peristaltic-pumps.html
dir.indiamart.com/impcat/sintex-water-tanks/storage-capacity-1000-l-q15271150/
dir.indiamart.com/impcat/sms-alert-module.html
eshop.kirloskarpumps.com/openwell-submersible-pump-kosi-0520-0-5-hp-single-p
hase-210-volts-size-25-mm-x-25-mm.html
freightfarms.com/blog/labor-requirements-for-the-greenery
greenfinindia.com/product/sun-sun-cuv-8110-55w-2-110uv-sterilizer-for-ponds-11
0-w-uv/
hrsoftwarebangalore.com/minimum-wages-karnataka/
link.springer.com/article/10.1023/A:1008798710325
link.springer.com/article/10.1186/s40538-025-00817-8
microworms.in/buy-sobo-submersible-water-pump-india
onyxaqua.com/product/sobo-wp-2880-aquarium-power-head-25-watts/
university.upstartfarmers.com/blog/best-crops-for-raft-systems
university.upstartfarmers.com/blog/sizing-a-pump-hydroponics-aquaponics
us.towergarden.com/products/pump
aadyahydro.in/aeroponic-growing-tower.html
amazon.in/Sobo-Aquarium-Submersible-Pump-WP-8000/dp/B09SG9GFLT
amazon.in/Sobo-Aquarium-Submersible-WP-5000-Power/dp/B09SG7ZN7V
amazon.in/Sobo-WP-WP-6000-Power-Submersible/dp/B0CNCNFMFLF
biorxiv.org/content/10.1101/2021.01.01.425018v1.full
citygreens.ai/blogs/post/cost-hydroponic-nft-farm-india
climatestotravel.com/climate/india/bangalore
countryandpolitics.in/2025/02/20/bescom-appeals-to-farmers-not-to-use-single-ph
ase-pump-sets-in-irrigation-feeders-at-night/
ctahr.hawaii.edu/hawaii/downloads/three_non-circulating_hydroponic_methods_fo
r_growing_lettuce.pdf
ctahr.hawaii.edu/oc/freepubs/pdf/VC-1.pdf
extensionjournal.com/article/view/1956/8-5-115
flipkart.com/sobo-submersible-pump-wp-7000-105w-f-max-5500-l-h-water-lifting-
pond-fountain-filter-sump-systems-foodie-puppies-aquarium/p/it...
indiamart.com/integratedhydroponicsindia/net-pots.html
indiamart.com/proddetail/16mm-online-agriculture-drip-irrigation-lateral-pipe-116
05459130.html
indiamart.com/proddetail/2-inch-net-pots-premium-round-hydroponic-net-pot-153
27478848.html
indiamart.com/proddetail/200-plants-dutch-bucket-hydroponic-system-22902490
173.html
indiamart.com/proddetail/aeroponic-tower-lotus-60-72-and-90-planter-commercia
l-setups-2856842162497.html
indiamart.com/proddetail/digital-adjustable-cyclic-timer-on-off-timer-2353106007
3.html
indiamart.com/proddetail/ebb-and-flow-hydroponic-system-21708383562.html
indiamart.com/proddetail/ebb-and-flow-nursery-table-25108426888.html
indiamart.com/proddetail/floating-raft-system-hydroponic-services-9930551188.h
tml
indiamart.com/proddetail/hdpe-pond-liner-sheets-3703454991.html
indiamart.com/proddetail/hydrogen-peroxide-50-21766675248.html
indiamart.com/proddetail/hydrogen-peroxide-50-9238090348.html
indiamart.com/proddetail/resun-lp-100-air-pump-22362969691.html
indiamart.com/proddetail/rhino-mat-hdpe-pond-liner-500-micron-with-isi-mark-34
27571848.html
indiamart.com/proddetail/sobo-submersible-aquarium-water-pump-wp-series-for-
aquarium-and-pond-27222733748.html
indiamart.com/proddetail/sobo-wp-8000-aquarium-submersible-pump-285222739
2973.html
indiamart.com/proddetail/upvc-nft-channel-100-x-50-mm-23992061473.html
indiamart.com/proddetail/urban-farm-dutch-bucket-hydroponics-18634443862.ht
ml
indiamart.com/proddetail/we-hydroponic-raft-xps-sheet-extruded-polystyrene-she
et-dwc-system-23258316333.html
lnbatteries.com/products/luminous-inverter-battery-combo-icon-1100-pure-sine-w
ave-inverter-rc-18000-150-ah-tall-tubular-battery-white-stand...
mdpi.com/1660-4601/12/6/6879
moglix.com/brands/kirloskar/pumps-motors/128000000
naturehydro.com/news/whats-the-best-flow-rate-for-an-nft-system
nobroker.in/blog/sintex-water-tank-price/
sciencedirect.com/science/article/pii/S0304423824003601
technode.co.in/power-sense-alert
uasbangalore.edu.in/wp-content/uploads/2024/12/3.pdf
whitecorals.com/en/dry-goods-/pumps-and-plumbing/booster-pumps/jecod-jebao-
dcp-pumps